Creating and Running Migrations

A migration is simply a Ruby source file in your application's `db/migrate` directory. Each migration file's name starts with a number of digits (typically fourteen) and an underscore. Those digits are the key to migrations, because they define the sequence in which the migrations are applied—they're the individual migration's version number.

The version number is the Coordinated Universal Time (UTC) timestamp at the time the migration was created. These numbers contain the four-digit year, followed by two digits each for the month, day, hour, minute, and second, all based on the mean solar time at the Royal Observatory in Greenwich, London. Because migrations tend to be created relatively infrequently and the accuracy is recorded down to the second, the chances of any two people getting the same timestamp is vanishingly small. And the benefit of having timestamps that can be deterministically ordered far outweighs the miniscule risk of this occurring.

Here's what the `db/migrate` directory of our Depot application looks like:

```
depot> ls db/migrate
20221207000001_create_products.rb
20221207000002_create_carts.rb
20221207000003_create_line_items.rb
20221207000004_add_quantity_to_line_items.rb
20221207000005_combine_items_in_cart.rb
20221207000006_create_orders.rb
20221207000007_add_order_id_to_line_item.rb
20221207000008_create_users.rb
```

Although you could create these migration files by hand, it's easier (and less error prone) to use a generator. As we saw when we created the Depot application, two generators create migration files:

- The *model* generator creates a migration to in turn create the table associated with the model (unless you specify the `--skip-migration` option). As the example that follows shows, creating a model called `discount` also